On rings of integers

Let K be a number field with ring of integers $\mathcal{O}_K$.

1 Hermitian line bundles over rings of integers

Definition 1. A *Hermitian line bundle* $\overline{\mathcal{L}}$ over $\operatorname{Spec}\mathcal{O}_K$ is a pair $(\mathcal{L}, (\|\cdot\|_\sigma)_{\sigma: K\to\mathbb{C}})$ where

- $\mathcal{L}$ is a line bundle over $\operatorname{Spec}\mathcal{O}_K$, i.e., a finitely generated projective $\mathcal{O}_K$-module of rank 1.
- For each embedding $\sigma : K \to \mathbb{C}$, $\|\cdot\|_\sigma$ is a hermitian metric on the complex vector space $\mathcal{L}_\sigma := \mathcal{L} \otimes_{\mathcal{O}_K,\sigma} \mathbb{C}$.

Example 2. Let $K = \mathbb{Q}(\sqrt{-5})$ with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. The class number of K is 2, and the ideal $\mathfrak{p} = (2, 1+\sqrt{-5})$ is a non-principal ideal in $\mathcal{O}_K$.

Yang:

Example 3. Let $K = \mathbb{Q}(\sqrt{10})$ with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{10}]$. The class number of K is 2, and the ideal $\mathfrak{p} = (2, \sqrt{10})$ is a non-principal ideal in $\mathcal{O}_K$.

Yang: To be continued.

Definition 4. Let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\operatorname{Spec}\mathcal{O}_K$. The *degree* of $\overline{\mathcal{L}}$ is defined as

$$\deg\overline{\mathcal{L}} := \log\#(\mathcal{L}/\mathcal{O}_K s) - \sum_{\sigma: K\hookrightarrow\mathbb{C}} \log\|s\|_\sigma,$$

where s is any non-zero section of $\mathcal{L}$. Yang: To be revised

Definition 5. An *arithmetic divisor* on $\operatorname{Spec}\mathcal{O}_K$ is a pair (D, g), where

- D is a Weil divisor on $\operatorname{Spec}\mathcal{O}_K$, i.e., a formal sum of closed points with integer coefficients;
- $g = (g_\sigma)_{\sigma: K\hookrightarrow\mathbb{C}}$ is a collection of Green's functions $g_\sigma : (\operatorname{Spec}\mathcal{O}_K)_\sigma(\mathbb{C}) \setminus \{D\} \to \mathbb{R}$ for each embedding $\sigma : K \hookrightarrow \mathbb{C}$.

Yang: To be revised

Definition 6. Let (D, g) be an arithmetic divisor on $\operatorname{Spec}\mathcal{O}_K$. The *degree* of (D, g) is defined as

$$\deg(D,g) := \sum_{x\in\operatorname{Spec}\mathcal{O}_K} n_x \log N(x) + \sum_{\sigma: K\hookrightarrow\mathbb{C}} g_\sigma(x_\sigma),$$

where $D = \sum_x n_x x$ and $N(x)$ is the norm of the closed point x. Yang: To be revised

Date: January 31, 2026, **Author**: Tianle Yang, My Homepage

Proposition 7. There is a one-to-one correspondence between isomorphism classes of hermitian line bundles on $\operatorname{Spec} \mathcal{O}_K$ and arithmetic divisors on $\operatorname{Spec} \mathcal{O}_K$. Moreover, this correspondence preserves degrees, i.e., for a hermitian line bundle $\overline{\mathcal{L}}$ corresponding to an arithmetic divisor (D, g), we have

$$\widehat{\deg}\, \overline{\mathcal{L}} = \widehat{\deg}(D, g).$$

Yang: To be revised

Proposition 8. Let K be a number field with ring of integers $\mathcal{O}_K$. There is an exact sequence

$$0 \to \mu_K \to \mathcal{O}_K^\times \xrightarrow{\log_K} \mathbb{R}^r \xrightarrow{l} \widehat{\mathrm{Cl}}(\mathcal{O}_K) \to \mathrm{Cl}(\mathcal{O}_K) \to 0,$$

where μ_K is the group of roots of unity in K, $\widehat{\mathrm{Cl}}(\mathcal{O}_K)$ is the arithmetic class group of $\mathcal{O}_K$, and $\mathrm{Cl}(\mathcal{O}_K)$ is the usual class group of $\mathcal{O}_K$. Yang: To be revised

2 Geometry of numbers

Definition 9. Let $\overline{\mathcal{L}}$ be a hermitian line bundle over $\operatorname{Spec} \mathcal{O}_K$. The *successive minima* $\lambda_1(\overline{\mathcal{L}}), \dots, \lambda_r(\overline{\mathcal{L}})$ are defined as

$$\lambda_i(\overline{\mathcal{L}}) := \inf\{\lambda > 0 : \dim_K K\{s \in H^0(\operatorname{Spec} \mathcal{O}_K, \mathcal{L}) : \|s\|_\sigma \le \lambda \text{ for all } \sigma\} \ge i\}.$$

Yang: To be revised

3 Riemann-Roch

Definition 10. Let $\overline{\mathcal{L}}$ be a hermitian line bundle over $\operatorname{Spec} \mathcal{O}_K$. The *Euler-Minkowski characteristic* $\chi(\overline{\mathcal{L}})$ is defined as

$$\chi(\overline{\mathcal{L}}) := \log \frac{\mathrm{vol}(B(\overline{\mathcal{L}}))}{\mathrm{vol}(\mathcal{L}_{\mathbb{R}}/\mathcal{L})},$$

where $B(\overline{\mathcal{L}}) = \{s \in \mathcal{L}_{\mathbb{R}} : \|s\|_\sigma \le 1 \text{ for all } \sigma\}$ and the volumes are computed with respect to the Lebesgue measure. Yang: To be revised

Appendix